SUPPORTING ADULT GUIDANCE

Solving by substitution

Equations, Inequalities and Simultaneous Relationships · Lesson 7. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson develops the **substitution** method for simultaneous linear equations: express one variable from one equation and substitute it into the other, reducing the system to a single equation in one unknown. It gives the exact common solution the previous lesson located graphically, and includes the rearrange-first case.

Driving Question: *How do I find the meeting point without drawing?*

The mathematics, in plain terms

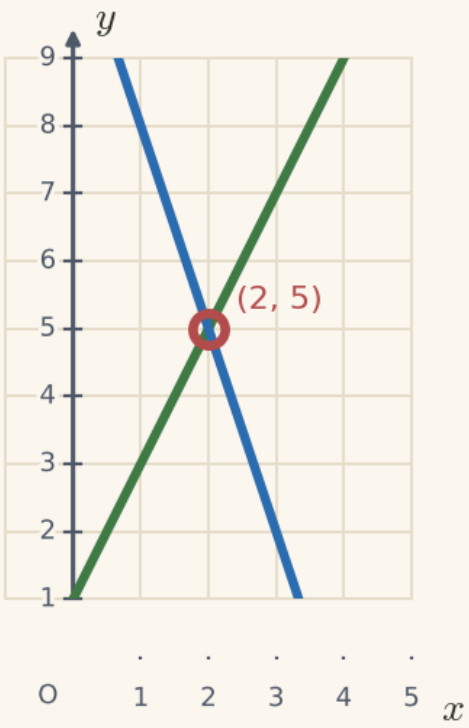
If one equation gives y (or x) explicitly, that expression may replace the variable in the other equation, since the two are equal — substitution preserves the system's solution. The result is one equation in one unknown; solving it and back-substituting yields the pair. When neither variable is isolated, rearranging one equation creates the needed subject. Checking substitutes the pair into both originals.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Substitute, solve, check

Solve $y = 2x + 1$ and $3x + y = 11$.

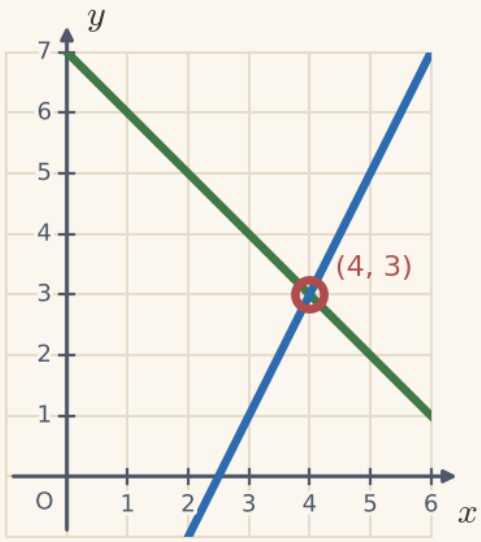


1. Substitute $(2x + 1)$ for y in the second equation.	$3x + (2x + 1) = 11$
2. Expand and collect.	$5x + 1 = 11$
3. Solve for x.	$x = 2$
4. Back-substitute for y, then check both.	$y = 5$; and $3\times2 + 5 = 11$ ✓

(2, 5) — substituting gives $5x + 1 = 11$, so $x = 2$ and $y = 5$; it satisfies both. This is the crossing.

Example 2 — Rearrange first

Solve $x + y = 7$ and $2x - y = 5$ — neither is in $y = \ldots$ form yet.



1. Rearrange the first to make y the subject.	$y = 7 - x$
2. Substitute into the second.	$2x - (7 - x) = 5$
3. Expand and collect, then solve.	$3x - 7 = 5$, so $x = 4$
4. Back-substitute and check.	$y = 3$; and $4 + 3 = 7$ ✓

(4, 3) — rearranging to $y = 7 - x$ lets substitution run as usual; $x = 4$, $y = 3$ satisfies both.

Substitute, collect, solve for x, back-substitute, check

Keep the sequence explicit: substitute the expression for y, expand and collect, solve for x, put x back to find y, check in both. The common slips are substituting into the same equation the expression came from (which gives 0 = 0), and forgetting to back-substitute for y. Reserve 'adding the equations' for the next lesson — here the move is to swap y for what it equals.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	See that y can be swapped for its expression, leaving one unknown.	~8 min

Movement	Work the full four-step substitution method, then check both.	~13 min
Reflection	Watch the algebraic answer land exactly on the graphical crossing.	~9 min
Creativity	Rearrange an equation first, then substitute.	~8 min
Rest	A short pause on 'equal things can stand in for each other'.	~2 min
Nutrition	Intellectual: an exact method, and reducing a problem to a familiar one.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Substituting back into the same equation the expression came from.
That just restates it (0 = 0). 'You took y = … from equation 1 — substitute it into equation 2, the other one.'

Solving for x but forgetting y.
A solution is a pair. 'You have x — now put it back into y = … to find y.'

Sign slips when substituting a subtraction, e.g. $2x - (7 - x)$.
Bracket the expression and expand carefully: $2x - (7 - x) = 2x - 7 + x$. 'What happens to each term inside when the minus goes in?'

Not checking in both equations.
'Put (x, y) into each original — do both come out true?' One check is not enough.

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: substituting a value or expression, rearranging to make a variable the subject, and solving a two-step equation. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner substitutes back into the wrong equation, or stops at x. The aim is a question that returns the method to them.

- "Which equation did y = … come from? Substitute it into the *other* one."
- "You have x. What do you still need — and how do you get y?"

- “When the minus goes into (7 – x), what happens to each term?”
- “Put your pair into both equations — do they each come out true?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Which variable should they isolate?

Whichever is easiest — one already written as y = …, or a variable with coefficient 1 so rearranging avoids fractions. Any correct choice reaches the same solution.

When is substitution better than elimination?

When one equation is already (or easily) in the form y = … or x = …. If both are in the form ax + by = c with tidy coefficients, elimination (next lesson) is often quicker. Both are valid.

Do the answers have to be whole numbers?

No — the stretch gives x = 0.5. Substitution handles fractional solutions exactly, which is one of its advantages over reading a graph.

How does this connect to Lesson 6?

Setting the two expressions for y equal (as in Lesson 6) is substitution in disguise. Here it becomes a reliable, general procedure that also gives y and copes with equations not written as y = ….

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.

Secure	Works through independently and explains why each step is true.
Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner’s thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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